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Linux commands

1)
```
┌──(kali㉿ankit)-[~]
└─$ pwd
/home/kali
```

pwd - Print working directory

2)
```
┌──(kali㉿ankit)-[~]
└─$ ls
ankit_practice  Desktop  Documents  Downloads  Music  my_session.txt  Pictures  Public  Templates  test.cxt  Videos
```

ls – List files and directories.

3)
```
┌──(kali㉿ankit)-[~]
└─$ cd
```

cd – change the current working directory in a command line interface.

4)
```
┌──(kali㉿ankit)-[~]
└─$ mkdir ankit_practice_2
```

mkdir – create a new directory.

5)
```
┌──(kali㉿ankit)-[~]
└─$ cd Desktop
```

cd Desktop – change the current directory to the desktop folder.

6)

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ touch test.cxt
```

Touch text.cxt – create an empty file or update the timestamp of an existing file.

7)

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ mv test.cxt final.txt
```

mv test.cxt final .txt – rename the file test.cxt to test.txt

8)

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ whoami
kali
```

Display the username of the currently logged-in user.

9)

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ uptime
 05:58:40 up  2:02,  1 user,  load average: 0.10, 0.17, 0.21
```

Uptime – shows how long the system has been running, along with the current load average.

10)

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ touch test.cxt
```

Touch test.cxt - create an empty file named test.cxt (or update its timestamp if it already exists).

11)

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ hostname
ankit
```

hostname – display the name of the of the current computer on the network

.

12)

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ head test.cxt
```

head test.cxt -display the first few lines of the file test.cxt (by default, the first 10 lines).

13)

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ tail test.cxt
```

tail.test.cxt-display the last few lines of the file test.cxt (by default, the last 10 lines).

14)

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ df -h
Filesystem      Size  Used Avail Use% Mounted on
udev            883M     0  883M   0% /dev
tmpfs           198M  972K  197M   1% /run
/dev/sda1        79G   16G   60G  21% /
```

df -h - shows disk space usage of file systems in a human-readable format (KB, MB, GB).

15)

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ cat test.cxt
```

Cat test.cxt - display the entire contents of the file test.cxt.

16)

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ free -m
               total        used        free      shared  buff/cache   available
Mem:            1971         922         692          23         520        1048
Swap:            953           0         953
```

Free -m - displays the system's memory usage (RAM) in megabytes.

17)

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:63:b0:05 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 71178sec preferred_lft 71178sec
    inet6 fd17:625c:f037:2:db7d:37ff:75bd:83ae/64 scope global dynamic noprefixroute
       valid_lft 86394sec preferred_lft 14394sec
    inet6 fe80::cff8:d8bd:201a:a254/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Ip a - display all IP addresses and network interfaces on the system.

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ ping google.com
PING google.com (142.250.70.110) 56(84) bytes of data.
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=1 ttl=255 time=8.49 ms
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=2 ttl=255 time=11.2 ms
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=3 ttl=255 time=9.16 ms
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=4 ttl=255 time=8.15 ms
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=5 ttl=255 time=8.58 ms
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=6 ttl=255 time=27.8 ms
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=7 ttl=255 time=10.2 ms
64 bytes from pnbomb-ac-in-f14.1e100.net (142.250.70.110): icmp_seq=8 ttl=255 time=8.93 ms
^C
--- google.com ping statistics ---
8 packets transmitted, 8 received, 0% packet loss, time 7010ms
rtt min/avg/max/mdev = 8.146/11.563/27.802/6.208 ms
```

Ping google.com - check network connectivity to Google by sending test packets and receiving replies.

19)

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ date
Wed Dec 31 10:09:03 AM EST 2025
```

date – display the current system date and time.

20)

```
┌──(kali㉿ankit)-[~/Desktop]
└─$ uptime
 10:20:52 up  6:24,  1 user,  load average: 0.36, 0.23, 0.14
```

uptime - shows how long the system has been running and the current load average.

21)

```
┌──(kali㉿ankit)-[~]
└─$ whois
Usage: whois [OPTION]... OBJECT...

-h HOST, --host HOST   connect to server HOST
-p PORT, --port PORT   connect to PORT
-I                     query whois.iana.org and follow its referral
-H                     hide legal disclaimers
      --verbose        explain what is being done
      --no-recursion   disable recursion from registry to registrar servers
      --help           display this help and exit
      --version        output version information and exit

These flags are supported by whois.ripe.net and some RIPE-like servers:
-l                     find the one level less specific match
-L                     find all levels less specific matches
-m                     find all one level more specific matches
-M                     find all levels of more specific matches
-c                     find the smallest match containing a mnt-irt attribute
-x                     exact match
-b                     return brief IP address ranges with abuse contact
-B                     turn off object filtering (show email addresses)
-G                     turn off grouping of associated objects
-d                     return DNS reverse delegation objects too
-i ATTR[,ATTR]...      do an inverse look-up for specified ATTRibutes
-T TYPE[,TYPE]...      only look for objects of TYPE
-K                     only primary keys are returned
-r                     turn off recursive look-ups for contact information
-R                     force to show local copy of the domain object even
                       if it contains referral
-a                     also search all the mirrored databases
-s SOURCE[,SOURCE]...  search the database mirrored from SOURCE
-g SOURCE:FIRST-LAST   find updates from SOURCE from serial FIRST to LAST
-t TYPE                request template for object of TYPE
-v TYPE                request verbose template for object of TYPE
-q [version|sources|types]  query specified server info
```

Whois- used to get registration and ownership information about a domain name or IP address.

22)

```
┌──(kali㉿ankit)-[~]
└─$ ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fd17:625c:f037:2:db7d:37ff:75bd:83ae  prefixlen 64  scopeid 0x0<global>
        inet6 fe80::cff8:d8bd:201a:a254  prefixlen 64  scopeid 0x20<link>
        ether 08:00:27:63:b0:05  txqueuelen 1000  (Ethernet)
        RX packets 339  bytes 36587 (35.7 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 433  bytes 43833 (42.8 KiB)
        TX errors 0  dropped 15 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
```

Ifconfig - display and configure network interface settings.

23)

```
┌──(kali㉿ankit)-[~]
└─$ man ls
```

```
LS(1)                                User Commands

NAME
       ls - list directory contents

SYNOPSIS
       ls [OPTION]... [FILE]...

DESCRIPTION
       List information about the FILEs (the current directory by default).  Sort entries alphabetically if none of -cftuvSUX nor --sort is specified.

       Mandatory arguments to long options are mandatory for short options too.

       -a, --all
              do not ignore entries starting with .

       -A, --almost-all
              do not list implied . and ..

       --author
              with -l, print the author of each file

       -b, --escape
              print C-style escapes for nongraphic characters

       --block-size=SIZE
              with -l, scale sizes by SIZE when printing them; e.g., '--block-size=M'; see SIZE format below

       -B, --ignore-backups
              do not list implied entries ending with ~

       -c     with -lt: sort by, and show, ctime (time of last change of file status information); with -l: show ctime and sort by name; otherwise: sort by ctime, newest first

       -C     list entries by columns

       --color[=WHEN]
              color the output WHEN; more info below

       -d, --directory
              list directories themselves, not their contents

       -D, --dired
              generate output designed for Emacs' dired mode

       -f     same as -a -U

       -F, --classify[=WHEN]
              append indicator (one of */=>@|) to entries WHEN

       --file-type
              likewise, except do not append '*'
 Manual page ls(1) line 1 (press h for help or q to quit)
```

man ls -display the manual (help documentation) for the ls command.

```
┌──(kali㉿ankit)-[~]
└─$ sudo su
sudo: unable to resolve host ankit: Name or service not known
[sudo] password for kali:
┌──(root㉿ankit)-[/home/kali]
└─#
```

24)

sudo su- switch to the superuser (root) account with administrative privileges.

25)
```
┌──(root㉿ankit)-[/home/kali]
└─# echo "hello everyone"
hello everyone
```

echo - display a line of text or the value of variables in the terminal.

26)
```
┌──(root㉿ankit)-[/home/kali]
└─# ps
    PID TTY          TIME CMD
 209269 pts/2    00:00:00 sudo
 209270 pts/2    00:00:00 su
 209280 pts/2    00:00:03 zsh
 213668 pts/2    00:00:00 ps
```

ps - display information about active (running) processes on the system.